

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Defined **DFT (Design for Testability)** strategies and developed pre-tapeout verification simulations for mixed-signal ATE platforms, aiming to improve testability and reduce production costs.
- Engineered **multi-site probe and test solutions** on high-performance mixed-signal ATE platforms, directly supporting the fast ramp of ultra-high-volume new products.
- Executed test development phases including silicon functionality debug, characterization, and yield analysis, optimizing solutions and managing offshore production release for new product introductions.
- Contributed to the continuous advancement of measurement methodologies through improvement initiatives within the wider Test Development community.
- Developed **Python scripts** for automating data extraction and initial analysis on test results, reducing manual processing time by approximately 30% for characterization reports.
- Collaborated with NP development and cross-functional teams to integrate and validate new hardware and software test solutions for leading-edge product portfolios.

SKILLS

Core Engineering – Analogue Electronics, Digital Electronics, DFT, Mixed-Signal ATE, Probe and Test Solutions, Silicon Functionality Debug, Characterization, Yield Analysis, Production Release.

Programming Languages – Python, Java, C, Visual Basic.

Software & Tools – Design Simulation tools, Jenkins, Git, Jira, Confluence, Linux, Splunk, Dynatrace.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker.

Databases & Scripting – SQL (PostgreSQL, MySQL, Oracle), Shell Scripting.

PROJECTS

ATE Platform Test Solution Development

- Designed and implemented a new **multi-site test solution** for a mixed-signal device on an ATE platform, enabling simultaneous testing of 8 units and increasing throughput by 50%.
- Developed **Python scripts** to automate the parsing of ATE test logs and generate preliminary yield reports, identifying common failure modes 2 days earlier in the development cycle.
- Collaborated with the hardware design team to integrate DFT features, improving test coverage by an estimated 15% and reducing debugging time for silicon anomalies.

Automated Test Data Analysis Framework

- Built a **Python-based framework** to ingest and analyze large volumes of test data from various product lines, facilitating faster characterization and yield analysis.
- Implemented data validation checks using **SQL queries** against a PostgreSQL database to ensure data integrity before analysis, catching approximately 20% of data input errors.
- Created automated reports summarizing key performance indicators and yield trends, presenting findings to the Test Development team weekly to inform optimization efforts.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)